

Hanjun Park

Department of Industrial, Manufacturing, & Systems Engineering
Texas Tech University, Box 43061, Lubbock, TX 79409-3061
✉ hanjpark@ttu.edu Lab webpage: <https://hanj321.github.io>

ACADEMIC APPOINTMENT

Assistant Professor <i>Texas Tech University</i> • Department of Industrial, Manufacturing, and Systems Engineering	09/2025–Present
Graduate Research Assistant <i>Occupational Ergonomics and Biomechanics Laboratories</i> • Faculty Advisors: Maury A. Nussbaum & Divya Srinivasan	2019–2025
Graduate Research Assistant <i>Life Enhancing Technology Laboratory</i> • Faculty Advisor: Woojin Park	2012–2014

EDUCATION

Ph.D., Industrial and Systems Engineering Virginia Tech, Blacksburg, VA • Graduate Certificate in Data Analytics	08/2025
M.S., Industrial Engineering Seoul National University, Seoul, Korea	08/2014
B.S., Industrial and Systems Engineering Georgia Institute of Technology, Atlanta, GA	12/2011

PUBLICATIONS

Peer-Reviewed Journal Publications (* corresponding-author)

- [J.8] Santos, F., Son, C.*, **Park, H.**, Xue, C., Yang, J., & Liang, N. (2026). Design, fabrication, and testing of an ergonomic intervention for a virtual reality headset using 3D printing. *Ergonomics*. Accepted
- [J.7] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2026). Adaptation to a whole-body powered exoskeleton: Human-exoskeleton coordination during load-handling tasks. *Annals of Biomedical Engineering*. 1-15. doi:10.1007/s10439-026-04025-9
- [J.6] **Park, H.***, & Nussbaum, M.A. (2025). Muscle synergy analysis during pseudo-static and dynamic overhead tasks with arm-support exoskeletons. *Journal of Biomechanics*, 113135. doi:10.1016/j.jbiomech.2025.113135

- [J.5] Ransing, V., Park, J., Ye, Y., **Park, H.**, Kim, S., Du, J., & Srinivasan, D.* (2025). Efficacy of virtual reality-based approach for users to understand the potential benefits and limitations of using exoskeletons. *Human Factors*, 67(11), 1136-1151. doi:10.1177/00187208251346627
- [J.4] **Park, H.**, Noll, A., Kim, S., & Nussbaum, M.A.* (2025). Passive arm-support and back-support exoskeletons have distinct phase-dependent effects on physical demands during cart pushing and pulling: An exploratory study. *Applied Ergonomics*, 126, 104510. doi:10.1016/j.apergo.2025.104510
- [J.3] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2023). A pilot study investigating motor adaptations when learning to walk with a whole-body powered exoskeleton. *Journal of Electromyography and Kinesiology*, 69, 102755. doi:10.1016/j.jelekin.2023.102755
- [J.2] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2022). Effects of using a whole-body powered exoskeleton during simulated occupational load-handling tasks: A pilot study. *Applied Ergonomics*, 98, 103589. doi:10.1016/j.apergo.2021.103589
- [J.1] **Park, H.**, Park, W.*, & Kim, Y. (2014). Manikin families representing obese airline passengers in the U.S. *Journal of Healthcare Engineering*, 5(4), 479-504. doi:10.1260/2040-2295.5.4.479

Journal Publications in Progress

- [U.1] **Park, H.***, & Nussbaum, M.A. Characterizing adaptation to using exoskeletons: A narrative review and assessment framework for occupational contexts. *Applied Ergonomics*. *Under Review*
- [U.2] Santos, F., Son, C.*, **Park, H.**, Xue, C., Yang, J., & Liang, N. Biomechanical effects of virtual reality headset support and task demands on the neck. *IIE Transactions on Occupational Ergonomics and Human Factors*. *In Revision*
- [U.3] **Park, H.***, & Nussbaum, M.A. Perceptual adaptations to back-support exoskeletons: evidence from measures of workload and discomfort. *In Preparation*
- [U.4] Lee, W., Kwon, S., **Park, H.***, Hwang, D.* When user language meets manufacturing standards: bridging the gap in 3D printing troubleshooting with domain-specific natural language processing. *In Preparation*
- [U.5] **Park, H.***, & Nussbaum, M.A. Motor adaptations to back-support exoskeletons: evidence from measures of muscle activity, kinematics, and kinetics. *In Preparation*

Peer-Reviewed Conference Proceedings

- [C.6] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D. (2022). Changes in kinematics and muscle activity when learning to use a whole-body powered exoskeleton for stationary load handling. *Proceedings of the 66th Human Factors and Ergonomics Society Annual Meeting*, 66(1), 273–274.

- [C.5] **Park, H.**, Lee, Y., Kim, S., Nussbaum, M.A., & Srinivasan, D. (2021). Gait kinematics when learning to use a whole-body powered exoskeleton. *Proceedings of the 65th Human Factors and Ergonomics Society Annual Meeting*, 65(1), 1367–1368.
- [C.4] **Park, H.**, Kim, S., Lawton, W., Nussbaum, M.A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual handling. *Proceedings of the 64th Human Factors and Ergonomics Society Annual Meeting*, 64(1), 888–889. - **Won Best Student Paper Award at the Occupational Ergonomics Technical Group**
- [C.3] Jung, J., **Park, H.**, Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2014). A review on interaction techniques in virtual environments. *Proceedings of the 2014 International Conference on Industrial Engineering and Operations Management* (pp. 1582-1590).
- [C.2] Beck, D., Hwang, D., Son, M., Jung, J., **Park, H.**, Park, J., & Park, W. (2013). A Review on Techniques for Evaluating Physical Stresses Associated with the Use of Various Input Devices. *Proceedings of the 2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea*.
- [C.1] Jung, J., **Park, H.**, Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2013). A Review on Object Manipulation Techniques in Virtual Environments and their Performance Metrics. *Proceedings of 2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea*.

HONORS & AWARDS

- Best Experimental Paper Award, OETG at HFES 2024
- Best Student Poster Presentation Award, ASSP Regional Conference 2023
- IISE Future Faculty Fellow (3F) Program Awardee 2023–2024
- NIOSH Training Fellow, Virginia Tech 2023–2025
 - Fellowship includes monthly stipend, tuition, and required fees
- ISE Graduate Travel Award (\$900), Virginia Tech 2022
- Best Student Paper Award, OETG at HFES 2020
- VT HFES Student Chapter Research Competition Award 2020
- Grado Department of ISE Fellowship, Virginia Tech 2018
 - Fellowship includes monthly stipend, tuition, and required fees
- Brain Korea 21 Scholarship, Seoul National University 2014
- Lecture & Research Scholarship, Seoul National University 2013

PRESENTATIONS

Peer-reviewed Conference Oral Presentations

- [O.10] Santos, F., & Son, C., **Park, H.**, (2026). A 3D-printed shoulder support to reduce neck strain during virtual reality device use. *30th Applied Ergonomics Conference, Arlington, TX*.
- [O.9] Santos, F., **Park, H.**, & Son, C. (2025). A 3D-printed shoulder support to reduce neck strain during prolonged virtual reality (VR) device use: pilot findings. *69th Human Factors and Ergonomics Society Annual Meeting, Chicago, IL*.

- [O.8] **Park, H.**, Ojelade A., Kim, S., & Nussbaum, M.A. (2024). Effects of using arm-support exoskeletons on muscle coordination during a pseudo-static overhead task: preliminary analysis of muscle synergy. *68th Human Factors and Ergonomics Society Annual Meeting, Pheonix, AZ.* - **Won Best Experimental Paper Award at the Occupational Ergonomics Technical Group.**
- [O.7] **Park, H.**, Noll, A., Kim, S., & Nussbaum, M.A. (2023). Effects of using passive back- and arm-support exoskeletons for cart pushing and pulling. *67th Human Factors and Ergonomics Society Annual Meeting, Washington, DC.*
- [O.6] Noll, A., Nussbaum, M.A., Kim, S., & **Park, H.** (2023). Reliability Analysis of Task Performance and Subjective Measures for Assessment of Occupational Exoskeletons. *67th Human Factors and Ergonomics Society Annual Meeting, Washington, DC.*
- [O.5] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D. (2022). Changes in kinematics and muscle activity when learning to use a whole-body powered exoskeleton for stationary load handling. *66th Human Factors and Ergonomics Society Annual Meeting, Baltimore, MD.*
- [O.4] **Park, H.**, Lee, Y., Kim, S., Nussbaum, M.A., & Srinivasan, D. (2021). Gait kinematics when learning to use a whole-body powered exoskeleton. *65th Human Factors and Ergonomics Society Annual Meeting, Baltimore, MD.*
- [O.3] **Park, H.**, Kim, S., Lawton, W., Nussbaum, M.A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual handling. *64th Human Factors and Ergonomics Society Annual Meeting.*
- [O.2] Beck, D., Hwang, D., Son, M., Jung, J., **Park, H.**, Park, J., & Park, W. (2013). A Review on Techniques for Evaluating Physical Stresses Associated with the Use of Various Input Devices. *2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea.*
- [O.1] Jung, J., **Park, H.**, Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2013). A Review on Object Manipulation Techniques in Virtual Environments and their Performance Metrics. *2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea.*

Poster Presentations

- [P.3] Santos, F., Son, C., **Park, H.**, Xue. C, Yang, J., & Liang, N. Design, fabrication, and testing of an ergonomic intervention for virtual reality headset using 3D printing. (2026). *WCOE Research Day, Texas Tech University, Lubbock, TX, February.*
- [P.2] **Park, H.**, Ojelade, A., Kim, S., & Nussbaum, M. A. (2023). Effects of different arm-support exoskeletons on muscle activation patterns during a pseudo-static overhead task - preliminary analysis of muscle synergy. *37th American Society of Safety Professionals (ASSP) Region VI Professional Development Conference (PDC) in Myrtle Beach, South Carolina, September.* - **Won Student Presentation Award**
- [P.1] **Park, H.**, Kim, S., Nussbaum, M. A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual material handling. *2020 Annual HFES/INFORMS Poster Competition, Virginia Tech, VA, October.* - **Won Student Poster Presentation Award**

Invited Seminars

- [S.3] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Psychological Sciences, Texas Tech University, October.*
- [S.2] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Industrial, Manufacturing, and Systems Engineering, Texas Tech University, April.*
- [S.1] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Industrial and Systems Engineering, Kennesaw State University, March.*

Invited Panel Talk

- [T.1] **Park, H.** (2025). From Student to Professor: Exploring Careers in Occupational Ergonomics. Panel Member. *Occupational Ergonomics Technical Group of HFES, September.*

TEACHING EXPERIENCE

Instructor of Record

Texas Tech, Dept. of Industrial, Manufacturing, & Systems Engineering

- IE 5304: Biomechanics & Work Physiology (n = 11) *Spring, 2026*

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 4624: Work Physiology (n = 36) *Fall, 2023*
- ISE 3614: Human Factors Engineering and Ergonomics (taught online, n = 7) *Summer, 2023*

Guest Lecturer

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 3624: Industrial Ergonomics *Spring, 2023*
“NIOSH Lifting Equation and Controls” (n = 100)
- ISE 4644: Occupational Safety and Hazards Control *Fall, 2018*
“Review on Hazard Analysis and Prevention” (n = 20)

Graduate Teaching Assistant

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 3624: Industrial Ergonomics *Spring, 2019*
- ISE 4624: Work Physiology *Spring, 2019*
- ISE 4644: Occupational Health and Hazards Control *Fall, 2018*

PROFESSIONAL EXPERIENCES

Software Engineer, Military duty *10/2014–03/2018*
Kornic Glory, Seoul, Korea

Research Intern *06/2011–08/2011*
LG Electronics, Seoul, Korea

ADVISING & MENTORING

Dissertation/Thesis Committee Member

- Felipe Santos, TTU (PhD, IMSE) Chair: Changwon Son (2025–Present) [J.8], [O.10], [O.9], [U.2]
- Daniel Leibman, TTU (PhD, Psychology) Chair: Heesun Choi (2025–Present)
- Armina Rahman, TTU (PhD, IMSE) Chair: Changwon Son (2025–Present)
- Jessica Lillian Dedeaux, TTU (PhD, IMSE) Chair: Changwon Son (2026–Present)

Research Mentoring

Voluntary Postdoc Position

- Leonardo Wei, Texas Tech (PhD, ISE) (01/2026–Present)

M.S. Students

- Delaney Sheppard, Texas Tech (BIOE) (02/2026–Present)
- Alex Noll, Virginia Tech (ISE) (08/2022–08/2024) [J.4], [C.8], [C.7]

Undergraduate Students

- Enriquez Bazoalto Kaylee, Virginia Tech (ISE) (05/2025-08/2025)

PROFESSIONAL SERVICE ACTIVITIES

Service to Profession

Session Chair, the 69th HFES Annual Meeting, Atlanta, GA 2025

Archival Journal

- Reviewer, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 2026–Present
- Reviewer, *IEEE Transactions on Human-Machine Systems* 2025–Present
- Reviewer, *IIEE Transactions on Occupational Ergonomics and Human Factors* 2023–Present
- Reviewer, *Human Factors and Ergonomics Society Proceedings* 2021–Present

Service to Institution

Texas Tech Regional Science Fair (K-12) Special Award Judge 2026

Texas Tech Regional Science Fair (K-12) Ribbon Judge 2026

Dean's Representative, Dissertation Defense of Yves Valentin
(MS in Human Factors Psychology, TTU Department of Psychological Sciences) 2026

TTU Graduate Academic Advisor 2025–Present

VT Chapter of the Human Factors and Ergonomics Society (HFES)

- Officer: *Webmaster* 2020–2022

National Biomechanics Day

- *Led lab tours and biomechanics activities for local high school students* 2020–2025

VT Center for the Enhancement of Engineering Diversity (CEED) TechGirls Summer Camp

- *Led hands-on biomechanics activities* 2019

VT College of Engineering Alumni Outreach Event

- *Led lab tours for alumni* 2019

PROFESSIONAL AFFILIATIONS

Member, Institute of Industrial and Systems Engineers (IISE)	<i>2023–Present</i>
Member, Human Factors and Ergonomics Society (HFES)	<i>2020–Present</i>
Steering Committee member, K-Human Factors and Ergonomics Society (K-HFES)	<i>2021–Present</i>

REMARK

Dual Citizenship: U.S.A. and Republic of Korea